

module - 1

1. Square pattern

logic

- Print the same number of stars in every row
- Use nested loop where the outer loop prints row and inner loop print columns
- Each row contain n stars
- This forms a square of size $n \times n$

Approach

- Take the size n
- Run the outer loop from 1 to n
- For each row, run the inner loop from 1 to n
- Print $*$ in every iteration of the inner loop
- Move to the next line after each row

Dry run

Row

1

2

3

4

5

Stars printed

2. Hollow Square pattern

Logic

- Print stars only on the boundary of the square
- The boundaries include first, last, row and first and last column.
- Print spaces inside the square
- This create a hollow square

Approach

- Take the size n
- Use nested loop for rows and columns
- If the current position is on the boundary, print *
- otherwise, print a space.
- Repeat until all rows are completed.

Dry run

Row

1

2

3

4

5

output

* * * * *

* *

* *

* *

* * * * *

3. Triangle pattern

Logic

- Print stars equal to the current row number
- The number of stars increases by one in every row
- Nested loops are used to control rows and stars
- This forms a right-angled triangle

Approach

- Take the size n
- Run the outer loop from 1 to n
- Run the inner loop from 1 to current row number
- Print one star in each iteration
- Move to the next line after each row

Dry run

Row	star printed column
1	*
2	* *
3	* * *
4	* * * *
5	* * * * *

4. Inverted triangle pattern

Logic

- Start by printing n stars in first row
- Decreases the number of stars by one in every next row
- Continue until only one star is printed
- This forms an inverted right-angled triangle

Approach

- Take the size n
- Run the outer loop from 1 to n
- Run the inner loop from 1 to $n - \text{current row} + 1$
- Print one star in each iteration
- Move to the next line after each row.

Dry run

Row

1

2

3

4

5

Stars printed

**

*

5. Pyramid Pattern

Logic

- Print Spaces before the Stars to align the pattern at the centre
- The Spaces decrease and Stars increase in each row
- This forms a Centred Pyramid.

Approach

- run the outer loop for rows
- Print $n-i$ Spaces
- Print $(2*i - 1)$ Stars

Dry Run

Row

1

2

3

4

5

Output.

*

* * *

* * * * *

* * * * * * *

* * * * * * * *

6. Diamond pattern

Logic:-

- Combine a pyramid and an inverted pyramid
- First increase stars, then decrease them
- Adjust spaces accordingly to keep the shape centered

Approach:-

- Print the upper pyramid
- Print the lower inverted pyramid
- Use spaces before stars in both parts.
- Continue until the diamond is completed

Dry run

Row

1

2

3

4

5

6

7

8

9

Output

*

* * *

* * * * *

* * * * * *

* * * * * *

* * * * *

* * * *

* * *

*

7. Half Diamond pattern

Logic

- First, increase the number of stars in each row
- After reaching the maximum, decreases the stars by one in each row.
- No spaces are required
- This forms a half diamond.

Approach

- Print an increasing triangle
- print a decreasing triangle
- combine ~~the~~ both parts to form the half diamond
- Repeat until the pattern is complete.

Dry run

Row

Output

1

*

2

**

3

4

5

6

**

7

*

8

**

9

*

8 Half diamond inverted pattern

Logic

- Start with the maximum no of stars
- Decrease one star in each row
- continue untill only one star remains
- This forms an inverted half diamond

Approach

- Take the size n
- Run the outer loop for rows
- Print $(n - \text{row} + 1)$ stars in each row

Dry run

Row

1
2
3
4
5

Output

x x x x x
x x x x
x x x
x x
x

Number patterns

1 Square pattern

Logic

- Print the same number in every row and column
- The value remains constant throughout the pattern
- This forms a number square.

Approach

- Take the size n
- Run the outer loop for Row
- Run the inner loop for Column.

Dry run

Row

output

1

1 1 1 1

2

1 1 1 1

3

1 1 1 1

4

1 1 1 1

5

1 1 1 1

2 Incrementing number pattern

Logic

- Print numbers from 1 to n in every row
- Each row contains the same ~~no~~ sequence of numbers
- This forms an incrementing number square.

Approach

- Run the outer loop for rows
- Run the inner loop from 1 to n
- Print the current column number

Dry run

Row

Column

1

1 2 3 4 5

2

1 2 3 4 5

3

1 2 3 4 5

4

1 2 3 4 5

5

1 2 3 4 5

3. Right triangle Number pattern

logic

- Print numbers from 1 upto current row number
- Each row adds one more number than the previous row
- This forms a right triangle

approach

- Run the outer loop for loops
- Run the inner loop from 1 to current row
- Print the current number

Dry Run

Row	output
1	1
2	1 2
3	1 2 3
4	1 2 3 4
5	1 2 3 4 5

4 Inverted Diamond pattern

logic

- Start with five repeated number
- Decrease one number in each row while increasing the printed value.
- This forms an inverted number pattern.

Approach

- Start printing from 3
- Print decreasing occurrence in each row
- Increase the number after every row

Dry run

Row

1

2

3

4

5

output

33333

4444

555

66

7

5 Sandwich number pattern

logic

- Print repeated numbers which decreasing towards the middle
- After the middle row, increase the repetitions again
- Created a sandwich - pattern.

Approach

- Print the upper half with decreasing repetitions
- Print the middle row
- Print the lower half with increasing repetitions.

Dry run

row

1
2
3
4
5

output

3 3 3 3 3
4 4 4
5
4 4 4
3 3 3 3 3

6. Incrementing Diamond number pattern

logic

- Increase repetitions in the upper half
- Decrease repetitions in the lower half.

Approach

- Print Increasing pattern, then decreasing pattern.

Dry run

Row

output

1

3

2

4 4

3

5 5 5

4

6 6 6 6

5

7 7 7 7 7

6

6 6 6 6

7

5 5 5

8

4 4

9

3.